



Dissertation/Technical report writing

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7 core principles (7C)

- **Clear**: Use simple words, avoid confusing jargon, and state your main point right away.
- **Concise**: Cut extra words and unneeded fluff to respect the reader's time.
- **Concrete**: Use exact numbers, specific names, and clear facts instead of vague statements.
- **Correct**: Check your grammar, spelling, facts, and technical steps so the text is accurate.
- **Coherent**: Put your thoughts in a logical order using headings, lists, and smooth transitions.
- **Complete**: Give all the details, steps, and answers the user needs to finish the task.
- **Courteous**: Use a polite tone and clean formatting that makes the text easy and friendly to read.

Dissertation/project format

- HKU Dissertation format:
  https://gentext.ai/guides/en/university-of-hong-kong-thesis-format/

Project format:

- Around 6 pages in length (excluding any essential Appendices).
- Prepare using a suitable word processing package (e.g., Microsoft Word)
- A4-size sheets
- Font size: 12 "Times New Roman"
- 1.5 line spacing
- Use SI units
- Margin: 25 mm



Structure

- Front matter (preliminaries)
- Main sections
- Back matter (end matter)

Front matter arrangement

- Title of the dissertation, author's name, and date of submission
- Abstract (not exceeding one page, 500 words)
- Acknowledgements (acknowledging people helping you collect data or edit your report)
- Table of contents
- List of notations
- List of figures
- List of tables

Back matter arrangement

- AI declaration statement
- Appendix: Extra charts, maps, or data tables.
- Bibliography: A list of source books and articles.
- Index: An alphabetical list of topics and page numbers.
- Glossary: A list of hard words and their meanings.



Abstract

- Aims
- Contributions
- Main Approaches …
- Something that one can/should remember after reading once

Main section arrangement

- Introduction
- Related Works (Literature review)
- For each main chapter
  - Problem definition and formulation
  - Data (if relevant)
  - Methodology (e.g., data collection method, solution method)
  - Results and discussion
- Conclusions



Introduction

- Technical trends (with citations)
- Applications (with citations)
- User scenarios (with citations)
- General related works (with citations)
- Specific related works (with citations)
- My goal
- My achievement (contribution/value)
- Outline



Related works

- Generally related
- Specifically related
- Highly related (which we will compare to)
- Classification of related works
- Research gaps



Problem definition and formulation

- Statement of the problem to be solved
- Use of symbols, notations, and equations is very important.
- Objective: such as maximize/minimize something
- Evaluation metrics (such as speed, network throughput, power consumption, etc.)



Conclusions

- What you have done (being proud of) (Summary)
- Main findings/recommendations
- How come it is better (limitations)
- Future directions

Project writing – One possible outline

- Abstract
- Introduction
  - Importance, application, background, objectives, scope, value/contribution, outline
- Literature review
  - Point out the need for this research
- Methodology
- Results and discussion
- Conclusion
  - Summary, major findings, recommendations, limitations, and future studies
- AI declaration statement
- References

Project writing – Another Possible outline

- Abstract
- Introduction
  - Importance, application, background, literature, **research gaps**, objectives, scope, outline
- Methodology
- Results and discussion
- Conclusion
  - Summary, major findings, recommendations, limitations, and future studies
- AI declaration statement
- References

Plagiarism

- If your dissertation/report contains part of another person's book, journal paper, article, or even another student's project report or dissertation, including data, graphs, or computer programs, WITHOUT quoting the source, then you have committed plagiarism.

Plagiarism

- If you are found to be guilty of plagiarism, your report will not be accepted.

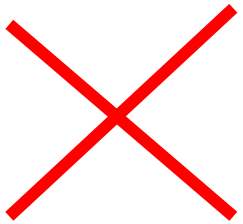
- Useful information from:

  Plagiarism and How to Avoid It – by D. Gardner

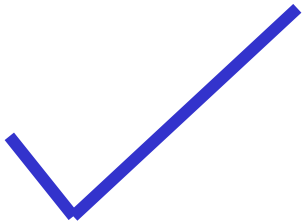
  https://caes.hku.hk/plagiarism/image/intro.pdf

Plagiarism vs Quotation/Citation

PLAGIARISM



QUOTATION



- The correct use of references and quotations is essential to the completion of a well-written project report.

Short Quotation

- Quotations of only a few words or a phrase should be indicated as shown below:

  In research aimed at investigating construction firms' marketing methods, Baker and Orsaah (1985) investigated "*how customers choose their contractor*" and found that low price, company financial standing and company reputation were the major factors.

Long Quotation

- Quotations of more than one sentence should be indented, single-spaced and written in italics as shown below:

Bromilow (1974) states that in his opinion:

> *"the most significant overall objectives in building operations are to define the design and specifications, price and timing of the proposed building, and, once they have been agreed by the client, to meet them."*

Quotations involving your own wording

- If the reference name naturally forms part of a sentence, then include it exactly as if no reference is being given. Follow the name by a space and the **full** year of publication, enclosed in parentheses.
  (Multiple authorship is dealt with in the same way.)
- Examples:
  - Carson (1970) argued that ...
  - The system developed by Brown and Smith (1986) is ...
    The declaration of human rights published by the United Nations (1948) was ...
  - On the other hand, Jones et al. (1988) have reported that ...

Quotations involving your own wording

- If the reference name does not form a natural part of the sentence, include both the name and the year, enclosed in parentheses, at an appropriate point in the sentence. (A comma can be placed between the name and the year, but this must be done **consistently.**)
- Examples:
  - Ronald Chan is very handsome (Szeto 2026).
  - An earlier paper (Carson 1970) argues that ...
  - The ARGA program (Brown and Smith 1986) is ...
  - The Universal Declaration of Human Rights (United Nations 1948) specified ...
  - On the other hand, it has been reported (Jones et al. 1983) that ..

Quotation style

- There are different proper citation styles. see:

  https://libguides.lib.hku.hk/c.php?g=766106&p=5501201

  https://student.unsw.edu.au/referencing

- Don’t mix different styles.

References (bibliographies)

- **"Full" details** of all published and unpublished books, articles, reports, and journal papers referred to in your project should be included in the reference list at the end of the main body of your report
- There are multiple referencing (and citing styles) even within the Harvard style
- Arranged alphabetically by author and then by year of publication (if the author has more than one publication).

References

- Book:

  Fellows, R., Langford, D., and Urry, S., 1983. *Construction Management in Practice*. Longman: Singapore.

- Book Chapter:

  McNally, M.G., 2008. The Activity-Based Approach. In Hensher, D.A. and Button, K.J. (eds) *Handbook of transport modelling,* pp. 55-74. Pergamon: New York.

- Report:

  MVA Asia Limited, 1993. *Travel Characteristics Survey Final Report.* Territory Transport Planning Division, Transport Department, Hong Kong Government.

References

- Journal paper:

  Leung, A.Y.T., 1987. A simple method for exponentially modulated random excitation. *Journal of Sound and Vibration*, Vol. 112, No. 2, pp. 273-282.

- Conference paper:

  Bates, J.J. and Roberts, M., 1983. Recent experiences with models fitted to stated preference data. *Proceedings of 11th PTRC Summer Annual Meeting*, University of Sussex, July 1983, England.

- Thesis:

  Stocks, R.K., 1984. *The Building Team: an Organisation of Organisations*, M.Sc. Thesis, The University of Hong Kong.

Using Endnotes?

References

- There are different proper ways to prepare a reference list. see:

https://libguides.lib.hku.hk/c.php?g=766106&p=5501201

MLA	Szeto, Wai Yuen, Yongzhong Wu, and Sin C. Ho. "An artificial bee colony algorithm for the capacitated vehicle routing problem." *European Journal of Operational Research* 215.1 (2011): 126-135.
APA	Szeto, W. Y., Wu, Y., & Ho, S. C. (2011). An artificial bee colony algorithm for the capacitated vehicle routing problem. *European Journal of Operational Research*, *215*(1), 126-135.
Chicago	Szeto, Wai Yuen, Yongzhong Wu, and Sin C. Ho. "An artificial bee colony algorithm for the capacitated vehicle routing problem." *European Journal of Operational Research* 215, no. 1 (2011): 126-135.
Harvard	Szeto, W.Y., Wu, Y. and Ho, S.C., 2011. An artificial bee colony algorithm for the capacitated vehicle routing problem. *European Journal of Operational Research*, *215*(1), pp.126-135.
Vancouver	Szeto WY, Wu Y, Ho SC. An artificial bee colony algorithm for the capacitated vehicle routing problem. European Journal of Operational Research. 2011 Nov 16;215(1):126-35.

- Don’t mix different styles.

What makes a good thesis: Foundations

- Demonstrate knowledge and identify its place within that field – **contribution to the literature** (in the introduction).
- Identify a **gap** in the existing scholarship – **have a good question** (in the introduction/literature review)
- Show the **significance** of this particular research for addressing that gap (in the introduction).
- Clearly inform readers of what to expect from the study, its **scope**, **aims** and **objectives** (in the introduction)

What makes a good thesis: Research design

- The *research design* must be **appropriate** to the objectives and address the questions identified in the foundations.

- *Methodology* should be **clearly described** so that the study can be replicated.

- Any innovative or *unconventional methodology*, or mode of analysis, needs to be **justified** to demonstrate that you were knowledgeably informed in choosing this approach.

What makes a good thesis: Research design

- *Results* should be **defensible**, and any possibility of bias or distortion in the data identified.

- The *results* should also be **presented appropriately** and **adequate interpretation** of the results carried out.

What makes a good thesis: Argument

- A good thesis has a **well-structured** and **focused argument**, **with links between diverse aspects** of the argument **clearly explained**.
- *Research materials* from authoritative sources will be **assimilated and integrated into the argument** of the thesis.
- A balance should be maintained between different sections, so that equal attention is given to the diverse elements that make up the whole.
- The significance of the results must be **related back to the existing literature**, **explaining convergences and divergences**.

What makes a good thesis: Argument

- *Discussion* of these should be **full** and **convincing**, so that each point is clearly articulated and conclusions are fore-grounded.

- *Conclusions* should be clearly linked to data and evidence, not based on speculation or opinion – **coherent**.

- The thesis should address its stated objectives, even if the outcome differs from what was anticipated – **have an answer to the question raised**.

What makes a good thesis: Writing

- **Well written** and **a pleasure to read**
- **Elegant**, **lucid** (expressed clearly; easy to understand), and **focused**
- Within the word limit, **concise** and **to the point**
- **Clearly and logically structured**
- **Thought provoking** and **interesting**
- Engaging for the reader, and will lead them clearly from point to point

What makes a good thesis: Writing

- Every paragraph should have a main idea or **topic sentence** (except in the abstract and conclusion)
- Every sentence within the paragraph should support the main idea, and every paragraph should support the main argument of the chapter/section.
- Every chapter must support the thesis as a whole.
- The quality of both the research and written argument should ideally present material that is of a **publishable standard**.
- Many thesis writing guides are available from the University and online. Take a look on them!

What makes a good thesis: Presentation

- The *presentation* of the thesis should be of a high standard with meticulous attention to detail, including spelling, grammar and punctuation – **good English** & **error free**!

- *Referencing* should follow the **recognized system** for your discipline and be **consistent**.

- The *format* and *presentation* of the document must **follow university requirements**.

AI declaration statement

- Declare how and which parts of their dissertations involve Generative AI

- Remarks
  - Don't trust all results from AI!
  - Read the following link for the responsible use of Generative AI in research at The University of Hong Kong:

    https://www.rss.hku.hk/integrity/generative-ai